

Dr Waseem Raza, PhD

Researcher & Engineering Educator | AI, Wireless Systems & Applied Machine Learning

AI for Wireless Networks | Reliable Learning Systems | Wearable Intelligence | Engineering Education

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Professional Summary

Researcher and engineering educator with a PhD in Electrical and Computer Engineering and experience spanning AI-enabled wireless networks, reliable machine learning, network resilience, propagation modeling, optimization, and wearable intelligence. Academic work combines analytical modeling, large-scale simulation, experimental validation, and data-driven methods, complemented by industry research at AT&T Labs and Futurewei and applied ML engineering in telecommunications. Research and teaching interests connect artificial intelligence with wireless and connected systems, dependable automation, and human-centered sensing.

Technical Expertise, Leadership, and Professional Strengths

Research Areas: AI for wireless networks, resilient and autonomous networks, propagation modeling, digital twins, RAN optimization, wearable intelligence, and data-efficient learning

Machine Learning: Deep learning, transfer learning, domain adaptation, adversarial learning, generative modeling, reinforcement learning, classification, time-series analysis, and model evaluation

Wireless Systems: LTE/4G, 5G-NR, NTN, mobility, radio access networks, propagation, outage management, network measurements, and service assurance

Research Methods: Experimental design, simulation, statistical analysis, cross-validation, robustness evaluation, reproducible workflows, and technical communication

Engineering & Data: Python, SQL, PyTorch, TensorFlow, scikit-learn, XGBoost, Pandas, NumPy, cloud analytics, data pipelines, and visualization

Teaching & Mentoring: Wireless communications, digital communications, networking, signals and systems, machine learning, programming, experimental methods, and student research development

Leadership and Professional Strengths

- **Research leadership:** Develop research questions into reproducible modeling, simulation, experimentation, validation, and publication workflows.
- **Teaching and mentoring:** Connect engineering fundamentals with applied examples and support students in technical problem solving, implementation, experimentation, and communication.
- **Industry translation:** Bridge academic methods with practical telecommunications problems through experience in university research and industry R&D environments.
- **Interdisciplinary collaboration:** Work across wireless communications, machine learning, data science, and wearable sensing while communicating results to technical and domain stakeholders.
- **Persistence and rigor:** Approach difficult research problems through systematic experimentation, careful validation, iterative refinement, and attention to reproducibility.
- **Continuous development:** Extend teaching and research capability through ongoing study in generative AI, agentic systems, cloud engineering, and emerging wireless technologies.

Professional Experience

Machine Learning Engineer

AT&T, contractor via Innovatix Inc.

Feb. 2025-Present

Bedminster, NJ

- Own end-to-end development, testing, and operational integration of an AI-driven diagnostic system for critical E911 service-assurance workflows, improving automated root-cause classification accuracy from 20% to 80%.
- Designed domain-adaptive neural models and reusable data-processing workflows that generalize across heterogeneous operational domains and reduce performance degradation caused by distribution shifts.
- Developed multimodal AI prototypes combining work-order text, location information, network context, and operational signals to support GenAI-assisted diagnostics for VoNR and 5G service assurance.
- Created model-validation, error-analysis, and failure-mode assessment procedures to compare performance across domains and support reliable deployment in high-impact operational environments.
- Partnered with network engineers, software teams, and public-safety stakeholders to translate operational requirements into validated AI workflows and production-oriented data assets.

Advanced-Degrees Intern

AT&T Labs

Jun. 2024-Aug. 2024

Austin, TX

- Analyzed 500K+ LTE/5G-NR diagnostic logs and applied ML methods to detect anomalies, reducing diagnostic time by 20%.
- Investigated 5G NR-NTN cell-selection behavior and improved decision algorithms by 15% through data-driven analysis and validation.
- Conducted structured analysis of large operational datasets and communicated technical findings to engineering teams supporting next-generation wireless systems.

Advanced Wireless Technology Development Intern

Futurewei Technologies

Sep. 2022-Jan. 2023

Schaumburg, IL

- Improved autoencoder-based channel-state-information compression by 25% using transfer learning and systematic experimentation on large wireless datasets.
- Conducted 50+ experiments to validate model performance, robustness, and generalization under changing data conditions.

- Collaborated with research and engineering stakeholders to translate modeling results into recommendations for next-generation communication systems.

Advanced-Degrees Intern
AT&T Labs

Jun. 2022-Aug. 2022
Bedminster, NJ

- Designed ML and statistical methods for analyzing users' timing in 5G standalone networks, improving analytical precision by 20%.
- Analyzed large-scale user trace data with SQL and visualization tools and communicated findings to senior technical leadership.
- Converted raw operational traces into interpretable performance indicators to support diagnosis and network optimization.

Graduate Research Assistant
AI4Networks, University of Oklahoma

Sep. 2019-Sep. 2024
Tulsa, OK

- Designed reusable Python simulation and data-generation platforms for large-scale experimentation, including a 3GPP-compliant network simulator and an AI-enabled 5G testbed.
- Built transfer-learning, adversarial-learning, GAN, autoencoder, and reinforcement-learning pipelines for sparse-data and cross-domain modeling.
- Developed robustness evaluation frameworks for models operating under distribution shifts, missing data, and limited labeled samples; led 100+ experiments from problem formulation through validation.
- Developed scalable self-healing and outage-management frameworks that improved modeled network resilience by 30%.
- Published peer-reviewed research, presented technical results internationally, and mentored junior researchers in ML implementation, experimental design, and technical writing.

Lecturer
Engineering and Technology Programs

- Taught and mentored undergraduate engineering students in communications, programming, and foundational engineering subjects.

Education

PhD, Electrical and Computer Engineering University of Oklahoma, Norman, Oklahoma, USA	09/2019-07/2025
MSc, Telecommunication Engineering University of Engineering and Technology, Taxila, Pakistan	08/2014-08/2016
BSc, Telecommunication Engineering University of Engineering and Technology, Taxila, Pakistan	11/2010-06/2014

Selected Publications

Real-World Medication Adherence Monitoring Using Hybrid Neural Networks and Smartwatches <i>HealthINF Accepted</i>	2026
On Multi-Parameter Optimization and Proactive Reliability in Cellular Networks <i>Journal article</i>	2025
An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin <i>IEEE PIMRC</i>	2024
Data-Driven Intelligent Outage Management for High Shadowing Environments in 5G&B Networks <i>IEEE SmartNets</i>	2024
Toward a Hybrid RF/Optical Lunar Communication System (LunarComm) <i>IEEE Network</i>	2022
Towards Positioning Error Impact Characterization and Minimization in User-Centric RAN <i>IEEE WCNC</i>	2022

Selected Projects

SyntheticNET: 3GPP-Compliant Network Simulation <i>Research Project</i>	2020-2024
• Python-based simulation environment for realistic mobility, propagation, and data-driven experimentation in cellular networks. <i>Tools & methods:</i> Python, 3GPP, Simulation, Wireless Networks	
TurboRAN Experimental Platform <i>Research Infrastructure</i>	2020-2024
• Multi-band, multi-tier experimental platform supporting evaluation of emerging AI-enabled radio-access-network solutions. <i>Tools & methods:</i> 5G Testbeds, RAN, Experiment Design, Network Measurement	
Wearable Intelligence for Health Monitoring <i>Research Project</i>	2024-2026
• Deep-learning methods for continuous recognition of medication and dietary activities from wrist-worn inertial sensors. <i>Tools & methods:</i> Wearable AI, Time Series, Deep Learning, Smartwatch Sensing	

Certifications and Continuing Education

1. Machine Learning — Stanford University / Coursera | Completed

Focus: Core supervised and unsupervised learning, model evaluation, optimization, and practical ML system development.

Tools & Skills: Supervised Learning, Unsupervised Learning, Model Evaluation, Optimization

2. AI for Telecommunications Specialization — [AI CERTs / Coursera](#) | Completed (3/3) courses | June 2026 | [Certificate](#)

Focus: Three-course specialization connecting AI foundations with telecommunications network optimization, security, and customer experience.

Tools & Skills: AI for Telecommunications, Network Optimization, Cybersecurity, Customer Experience, Internet of Things, Python, TensorFlow

3. 4G Network Fundamentals — [Institut Mines-Télécom / Coursera](#) | Completed (1/1) courses | August 2026 | [Certificate](#)

Focus: Foundational study of fourth-generation cellular networks.

Tools & Skills: 4G Networks, Cellular Networks, Telecommunications

4. Generative AI Fundamentals Specialization — [IBM / Coursera](#) | Completed (5/5) courses | July 2026 | [Certificate](#)

Focus: Five-course foundation in generative AI, prompt engineering, LLMs, and practical enterprise applications.

Tools & Skills: Generative AI, Prompt Engineering, Foundation Models, Responsible AI, IBM watsonx, Hugging Face, Business Applications

5. AWS Generative AI and AI Agents with Amazon Bedrock — [AWS/Coursera](#) | Completed (3/3) courses | 08/2026 | [Certificate](#)

Focus: Three-course professional certificate focused on developing, customizing, optimizing, and automating generative AI applications and intelligent agents with Amazon Bedrock.

Tools & Skills: Amazon Bedrock, Generative AI, AI Agents, Foundation Models, Knowledge Bases, Retrieval-Augmented Generation, LangChain, Amazon Q Developer, Model Customization, Prompt Engineering, Natural Language Processing, Text Generation, Summarization, Generative AI Application Development